

Vehicle Rental Management System

DBMS Mini Project Report

1. Introduction

The Vehicle Rental Management System is a database-driven web application designed to manage vehicles, customers and rental transactions. The application uses MySQL as the relational database and Python Flask as the web application framework.

2. Problem Statement

Manual vehicle rental management can lead to duplicate records, difficulty tracking vehicle availability, incorrect rental calculations and poor record management. The proposed system provides a centralized database and simple web interface.

3. Objectives

The main objectives are to store customer and vehicle information, manage rental bookings, track vehicle availability, calculate rental amounts, record returns and demonstrate DBMS concepts such as primary keys, foreign keys, joins and constraints.

4. ER Diagram

The main entities are Category, Vehicle, Customer and Rental. Category has a one-to-many relationship with Vehicle. Customer and Vehicle are connected through Rental records.

5. Database Schema

category(category_id, category_name)

customer(customer_id, name, email, phone)

vehicle(vehicle_id, category_id, registration_no, brand, model, vehicle_year, rental_price_per_day, status)

rental(rental_id, customer_id, vehicle_id, rental_date, return_date, total_amount, status)

6. Technologies Used

Python, Flask, MySQL, HTML5, CSS3, MySQL Connector/Python and Visual Studio Code.

7. SQL Queries and Features Implemented

The project uses CREATE DATABASE, CREATE TABLE, INSERT, SELECT, JOIN, UPDATE and foreign-key constraints. Rental status uses ENUM values including Booked, Ongoing, Completed, Cancelled and Returned. Returning a vehicle updates the rental record and changes the vehicle status back to Available.

8. Screenshots

Screenshots from the project development/testing process are included in the screenshots folder of the project ZIP. They can also be inserted into the final printed report.

9. Conclusion

The Vehicle Rental Management System demonstrates how a relational DBMS can be integrated with a web application to manage rental operations efficiently. The system provides a simple interface and maintains related data consistently.

10. Future Scope

Future versions can add authentication, online payment, customer registration forms, search and filtering, automated email notifications, invoice generation and a dashboard with rental statistics.